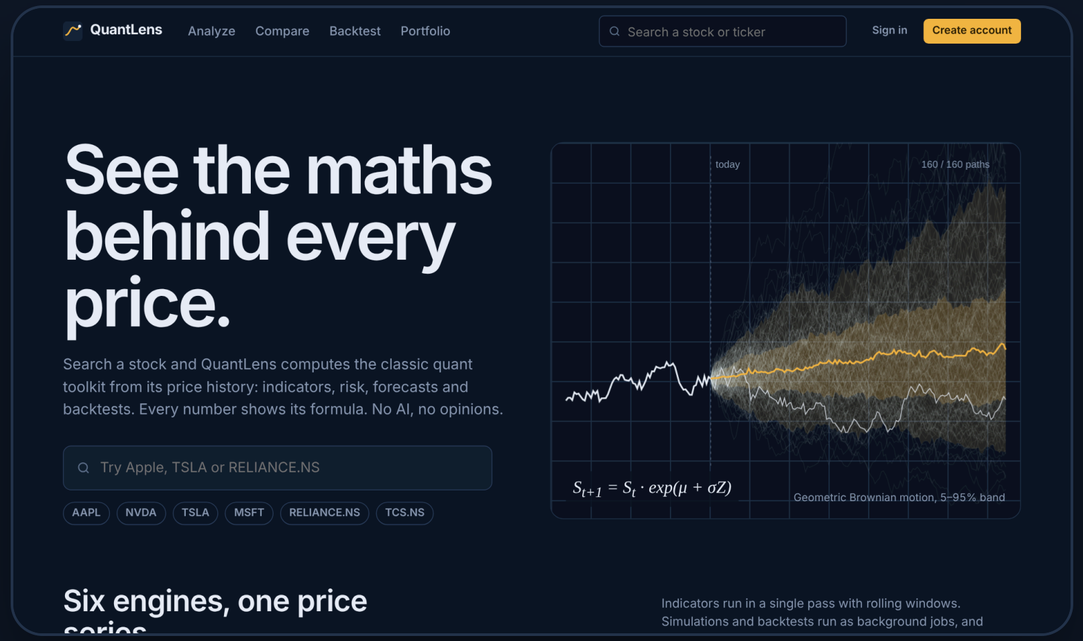
QuantLens

See the maths behind every price.

A mathematical stock analysis platform. Indicators, risk, forecasts and backtests. No AI, only math.



Team

Mohammed Raahil Ali · 160124748052

Shaik Mohammed Younus Saqib · 160124748059

Guide

Ms. Falak Naaz

Full Stack Web Development · 22CAC11N

Problem statement

Investors see a price, not the risk behind it

Hidden formulas

Free sites show indicators but never how a number was computed.

Risk is missing

Volatility, drawdown, VaR and beta need spreadsheets or paid terminals.

Testing needs code

Backtesting a simple rule means writing a program first.

API limits

Free market-data APIs block apps that refetch for every chart.

> *“Build a web app that fetches price history for any stock, computes indicators, risk, portfolio statistics, forecasts and backtests with transparent formulas, shows them on interactive charts, and stores each user’s data securely, within free API limits.”*

What is new

What makes QuantLens different

Need	Typical tools and projects	QuantLens
Explanation	Formulas hidden; AI gives opinions	Every number shows its formula; no AI
Forecast	One target price	1,000-path Monte Carlo fan with probabilities
Risk	Rarely shown	Volatility, Sharpe, drawdown, VaR, beta
Strategy test	Needs programming	No-code backtester vs buy-and-hold
Portfolio	Paid software	Live Markowitz efficient frontier
Engineering	Refetch on every click	Redis cache, BullMQ jobs, O(n) maths
Payments	Fake “Pro” button	Razorpay with server-side signature check

System design

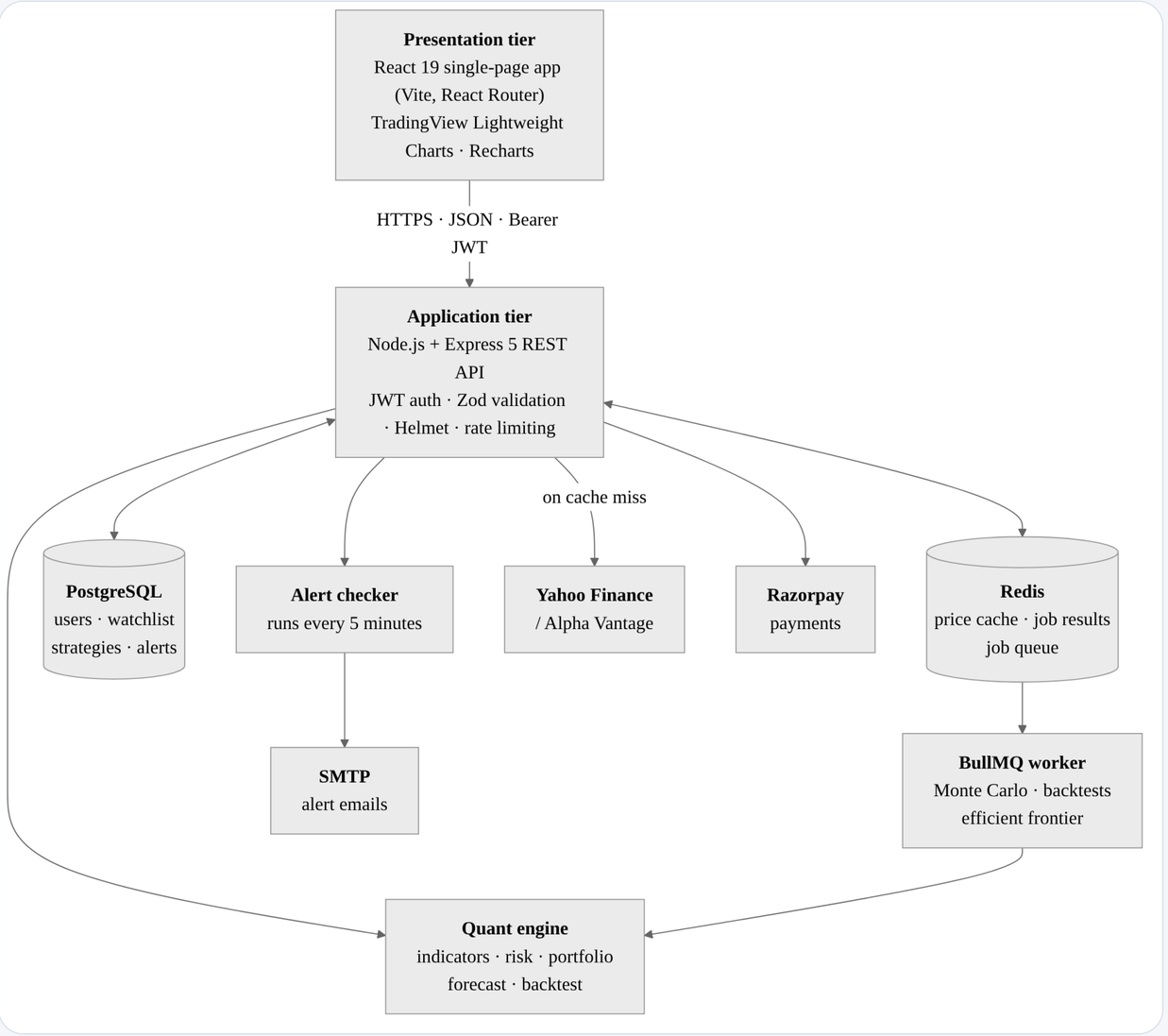
Three tiers, one queue

Presentation. React SPA talks only to the API over HTTPS with a JWT.

Application. Express answers fast requests; heavy jobs go to BullMQ workers.

Data. PostgreSQL keeps user data; Redis caches prices and results.

External. Yahoo Finance, Razorpay and SMTP.



Implementation

Four algorithms at the core

Indicators in one pass, O(n)

Rolling sums for SMA and Bollinger, recursive EMA and RSI, a monotonic deque for Stochastic highs and lows.

$$sum \mathrel{+}= P_t - P_{t-n}$$

Monte Carlo, 1,000 paths

Geometric Brownian motion calibrated on two years of log returns, seeded so results repeat.

$$\ln S_{t+1} = \ln S_t + \mu + \sigma Z$$

Markowitz efficient frontier

4,000 sampled long-only portfolios; the upper envelope is the frontier, plus max-Sharpe and min-volatility.

$$\mu p = w^T \mu,\ \sigma p = \sqrt{(w^T \Sigma w)}$$

Backtest without look-ahead

Signals on day t's close, trades at day t+1's open, with stop loss, take profit and fees.

$$signal(t) \rightarrow fill(open\ t+1)$$

Output · Stock analysis

Search a stock, get the full chart



Candlesticks

TradingView Lightweight Charts with volume

7 overlays

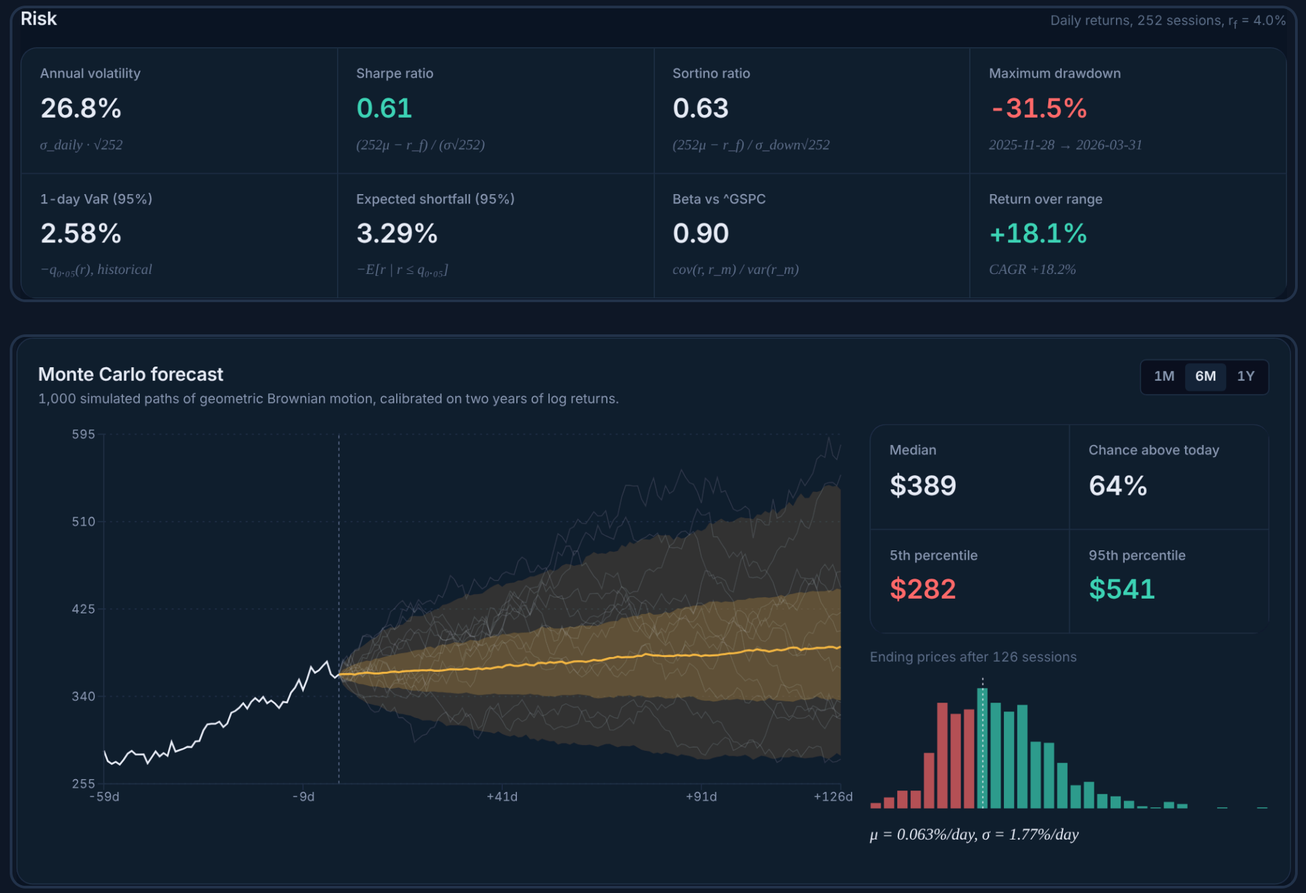
SMA 20/50/200, EMA 12/26, Bollinger, trendline

3 oscillators

Switch the lower pane: RSI, MACD or Stochastic

Output · Risk and forecast

Every number shows its formula



8 risk metrics

Volatility, Sharpe, Sortino, max drawdown, VaR, expected shortfall, beta, CAGR

Monte Carlo fan

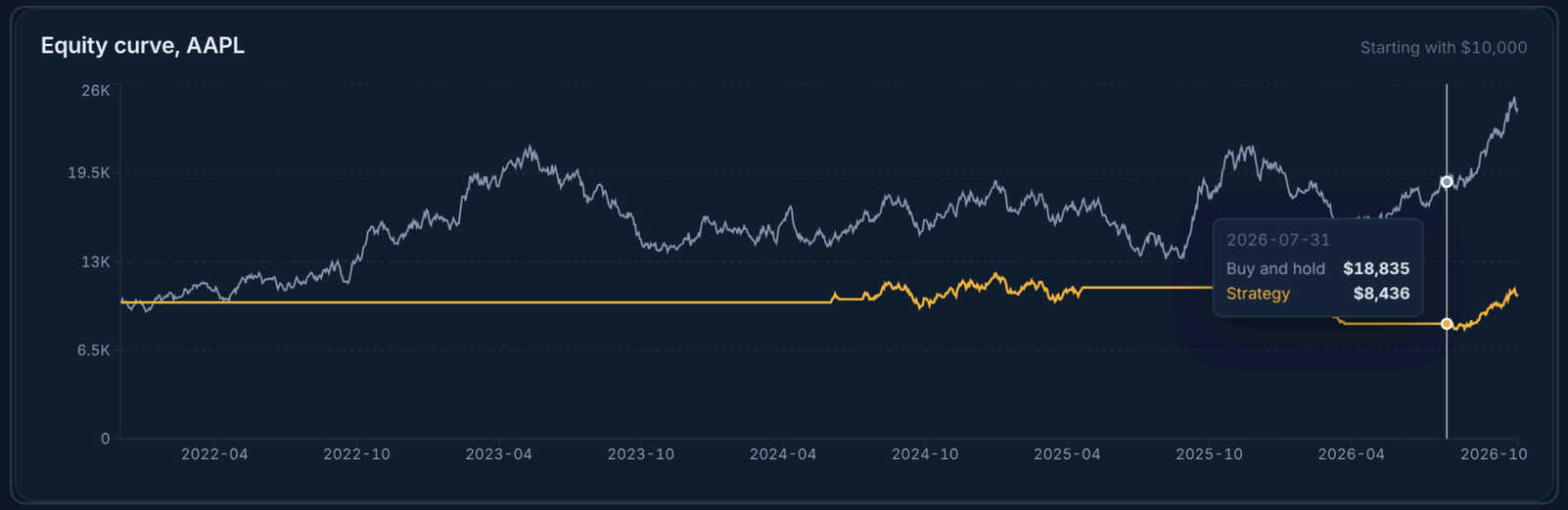
1,000 paths; 5–95% and 25–75% bands, median in amber

Ending prices

Histogram plus the chance of ending above today

Output · Strategy backtester (Pro)

Test a rule on five years of history



Strategy return	CAGR	Sharpe ratio	Max drawdown
+6.0%	+1.2%	-0.12	-33.7%
Buy and hold +142.3%	*Buy and hold +19.4%*	*Buy and hold 0.64*	*Buy and hold -38.6%*

Trades	Time in market
3	30%
Win rate 67%	*Avg trade -4.3%*

Rule: buy when SMA 50 crosses above SMA 200; sell on the reverse cross.

+6.0%

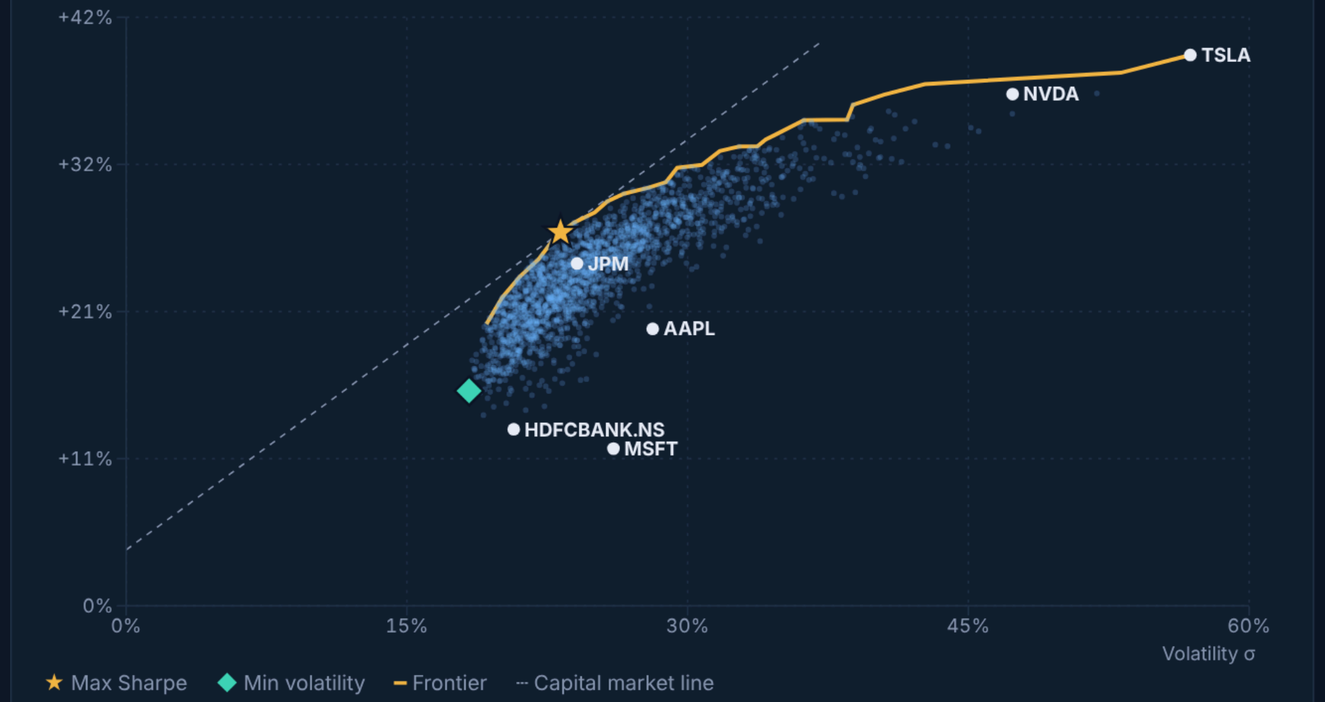
strategy return

+142.3%

buy and hold, same period

Output · Portfolio builder (Pro)

The efficient frontier, redrawn live



Weights

Max Sharpe | Min volatility

JPM	54.4%
NVDA	20.8%
AAPL	12.7%
TSLA	6.5%
HDFCBANK.NS	5.5%
MSFT	0.2%

Return	Volatility	Sharpe
+26.7%	23.2%	0.98

$$\sigma_p = \sqrt{(w^T \Sigma w)},\ \mu_p = w^T \mu,\ \Sigma w = 1,\ w \geq 0$$

Results and conclusion

15/15
automated math tests pass

1,000
Monte Carlo paths per forecast

4,000
portfolios sampled for the frontier

0
AI models: only maths

Next: live prices over WebSockets, more indicators, Razorpay subscriptions and cloud deployment with CI.

Thank you

Mohammed Raahil Ali · 160124748052
Shaik Mohammed Younus Saqib · 160124748059
Guided by Ms. Falak Naaz